Worksheet — Energy Conservation Simulation

Use with *Simulation - Energy conservation.html* · energy stores, conservation of energy and the flying-spring practical (Block 10)

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Name: _____ Class: _____ Date: _____

How to use this worksheet

Scan the QR code or open **school.taneski.com/physics/dp/20-energy-conservation/** in a browser (offline: *Simulation - Energy conservation.html*). Choose the example at the top: **Ramp & track · Pendulum · Flying spring · Drop**. Pick a preset in **Example** or set the values yourself (type a value and press Enter).

Press ▶️ Start; ◂ / ▸ step through the motion. The middle column shows the free-body diagram and the energy stores (switch **Energy chart** to Sankey). The dashed total line on *E*–*t* stays flat: energy is conserved.

Part 1 before the lab (≈ 15 min) · **Part 2** during the lab (≈ 20 min) · **Part 3** homework (≈ 30 min). $E_k = \frac{1}{2}mv^2$ · $\Delta E_p = mg\Delta h$ · $E_H = \frac{1}{2}kx^2$. g = 9.81 N kg^{-1}. Answers to 3 s.f. with units.

Scan to open

Part 1 — Pre-lab: predict, then test

Question	My prediction (before)	What the simulation showed	Physics explanation
1. 🎳 A heavy ball on a rope is held against the teacher's nose and released (not pushed). On the way back, does it hit the nose?			
2. Two smooth tracks drop 2.0 m: a steep curved one and a straight ramp. Which gives the higher speed at the bottom?			
3. A 0.50 kg and a 3.0 kg ball are dropped from the same height (no air). Which hits the ground faster?			
4. 🌀 A spring is stretched twice as far before it is released. How much higher does it fly?			
5. 🛹 A skater in a half-pipe with a little friction: where does the 'lost' mechanical energy go? Does the total energy change?			

6. Calculate the kinetic energy of a 0.50 kg ball moving at 4.0 m s^{-1}.

Working

7. Calculate the work needed to stretch a spring (k = 100 N m^{-1}) by 0.20 m.

Working

8. Calculate the GPE gained by a 3.0 kg bag lifted 2.0 m. Check: Drop → Retrieval: 3.0 kg bag.

Working

9. (exit question) A ball is dropped from 5.0 m. Use energy conservation to find its speed at the ground. Check: Drop → Exit question.

Working

Part 2 — During the lab

PASCO Smart Cart demo (lab 06): energy on a ramp. Simulation: Ramp & track → Class demo (0.500 kg, 1.00 m along a 6° ramp). Press ▶️ Start, step to the bottom and copy the values. The grey row has the class data (Capstone sample).

run	GPE at release / J	KE at the bottom / J	v at the bottom / m s^{-1}	√(2gh) / m s^{-1}	energy 'lost' / %	average friction / N
simulation						
class data	0.513	0.483	1.39			

10. Where did the 'lost' energy go? Draw a Sankey diagram to scale for the cart (switch Energy chart to Sankey to check).

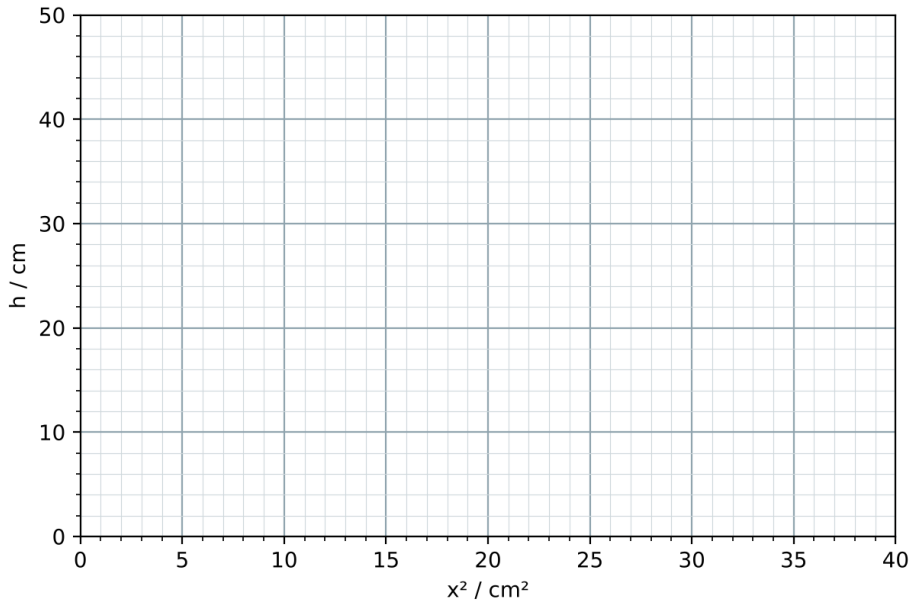
Sankey diagram (scale: _____ = 0.1 J)

11. 🎳 Pendulum to the nose: run the hook demo with air resistance on, then 'The same pendulum in a vacuum'. How far short of the nose does it stop each time? Explain.

Flying spring (Oxford practical). ⚠️ Safety glasses on. Stretch the spring by *x*, release it and measure the highest point *h* (three times). Simulation: Flying spring (10 g, k = 25 N m^{-1}); run 1.0–6.0 cm and ● Record each.

x / cm	x^2 / cm^2	h_1 / cm	h_2 / cm	h_3 / cm	mean h / cm	h (simulation) / cm
1.0						
2.0						
3.0						
4.0						
5.0						
6.0						

12. Plot h against x^2 and draw the best-fit line. Use the gradient to find k (gradient = k/2mg; for x in cm and h in cm, k = 2mg × gradient × 100).



Working

Part 3 — Homework

13. A frictionless roller coaster drops 20 m from rest. Calculate its speed at the bottom.

Working

14. A pendulum 2.0 m long is released from 30° to the vertical. Calculate the height it falls through and its speed at the bottom (no air resistance).

Working

15. Flying spring: 10 g, k = 25 N m^{-1}, stretched 4.0 cm. Calculate the height it reaches (no losses).

Working

16. A real spring loses 30 % of its elastic energy as vibration and sound when it is released. Will the k from your $h–x^2$ gradient be correct, too big or too small? By how much?

17. A 0.50 kg cart is released from 0.30 m and reaches the bottom at 2.2 m s^{-1}. What percentage of its GPE was dissipated? Where did it go?

Working

18. In the simulation, double the mass of the Smart Cart (friction μ unchanged). Does its speed at the bottom change? Explain using energy.